

TPA ENGINEERING CORPORATION

JTEKT

Head office :

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Messrs. WIA INDIA
KOREA

Date : Nov. 03, 2025

E S T I M A T E

No. TPAS20251103-01

ENQUIRER : NILL

REF. NO. : NILL

MACHINE : JTEKT CNC Production Anguler - Wheel slide Cylindrical Grinder
model GL5Ai-63

- as per attached brochure and/or our technical Specifications
No. VP-XXXXX

PRICE : F.O.B Busan port in Korea won
₩ 1,290,000,000 (VAT excluded)

DELIVERY : PLACE : WIA INDIA PUNE PLANT

TIME : 9 months after our receipt of your order note

PAYMENT : Usually

VALIDITY : 1 months after the date of this Estimate, or by the date of

We are much pleased in quoting you herein, and thanking for your valued orderin advance.

※ 한/일 합작 제작으로 원산지 증명서 발급 불가

Yours faithfully,



Jaehwan Park
President

TPA ENGINEERING CORPORATION

[A] TOYODA CNC Angular-Wheelslide Cylindrical Grinder
model 60M GL5Ai-63 (wheel dia. of 610 mm, wide width max 125mm Specification)

* Specially designed and tooled up for
 finish-grinding of OUTER RACE

* Our specification No.

1	<u>Basic Machine</u>			
1.1	Basic machine with GC70S1 system model 60M GL5Ai-63	1 set	¥	174,120,000
2	<u>Extra Equipment:</u>			
2.1.	Live spindle workhead	1 set	¥	4,500,000
2.2.	Footstock, automatic(NC) reposition type, manual taper adjustment type - NC stroke 190mm	1 set	¥	6,500,000
2.3.	Rotary diamond roll type wheel dresser, mounted on the rear of workhead	1 set	¥	6,500,000
2.4.	Hydraulic oil pump unit, included low oil level warning switch with galvanizing(only internal of tank)	1 set	¥	5,600,000
2.5.	Lubricating oil pump unit, included low oil level warning switch with galvanizing(only internal of tank)	1 set	¥	6,600,000
2.6.	Paper filter type coolant supply equipment with magnetic separator, oil skimmer and ,bed cleaning pump, piping for cleaning bed, coolant cooler and sludge box. - Sludge box material : SUS - Apply galvanizing and anti-slip to top plate - Unable to galvanizing for the internal of tank	1 set	¥	40,000,000
2.7.	Standard service tool kit with torque wrench	1 set	¥	480,000
2.8.	Manual overhead open/shut type full cover and totally-closed cover	1 set	¥	8,000,000
2.9.	Air blowing unit	1 set	¥	3,500,000
2.10.	Piping for lubricating to center hole, R,L-side	1 set	¥	700,000
2.11.	Addition of 6kinds of workpiece design fee with programing fee(total 7 kinds)	6 kinds	¥	18,000,000
2.12.	2 gripper type gantry loader - Fanuc 35i series(with 2years maintenance contract) - Gripper unit - Drive gear / Helical Gear type rack & pinion drive - Gantry oil pan with galvanizing - Excluded air blow	1 set	¥	120,000,000
Sub Total: [1]+[2]				¥ 394,500,000
3	<u>OPTIONAL ACCESSORIES</u>			
3.1.	Automatic electric in-process gauging system, with 1 pce. of "MARPOSS "-make table-mounted type hydraulically-operated gauging head and control unit , pipings, wirings and fittings (UNIMAR S26, AMP P7), including automatic lateral locator, and auto wheel balacer	1 set	¥	52,000,000

3.2.	Maser bar for M/C alignment	1 set	¥	850,000
3.3.	Master work(using by WIA work)	0 pcs.	Deleted	
3.4.	3-steps signal tower with buzzer	1 set	¥	300,000
3.5.	Cooler for electric control box	1 set	¥	3,000,000
3.6.	Local lighting unit	1 set	¥	250,000
3.7.	Lighting units inside control box	1 set	¥	250,000
3.8.	Jib crane with Elec. type hoist	1 set	¥	2,500,000
3.9.	Air gun & coolant gun	1 pce.	¥	200,000
3.10.	Mist collector attaching arrangement (excluding mist collector)	1 set	¥	1,000,000
3.11.	Terminal box between loader and machine	1 set	¥	700,000
3.12.	Wheel guard auto offset device	1 set	¥	3,000,000
3.13.	Master bar for M/C accuracy check			DELETED
3.14.	Extra application to machine and equipment for KCs standard - noise filter, high efficiency motor, etc.			DELETED
3.15.	X-axis retract function for power supply off	1 set	¥	3,000,000
3.16.	Wheel trolley—two wheel mount type			DELETED
3.17.	Wheel balance stand & arbor	1 set	¥	5,000,000
3.18.	UPS system for power failure preparation			DELETED
		Sub Total:	[3]	¥ 72,050,000
4	<u>Durable tool:</u>			
4.1	Grinding wheel flange(no spare)	4 sets	¥	23,200,000
4.2	Measuring head for auto locator(as spare)			DELETED
4.3	Measuring head for auto sizer(as spare)			DELETED
4.4	Feeler & contact for auto-locator(as spare)	1 sets	¥	1,500,000
4.5	Feeler & contact for auto-sizer(as spare)	1 sets	¥	3,000,000
4.6	Special type work driving center including driving plate (L-side only) (2 pcs. as spare)	3 pcs.	¥	6,000,000
4.7	Carbide-tipped tail stock center(R-side only)(2 pcs. as spare) - dead type center	3 pcs.	¥	1,500,000
4.8	Driving pin(for attaching drive center) - 2kinds X 2sets	4 sets	¥	1,000,000
4.9	Grinding wheel - 4kinds X 2sets	8 pcs.	¥	10,400,000
4.10.	Diamond roll(1 pce. as spare)	2 pcs.	¥	4,400,000
4.11.	Loader Jaw - 1kinds X 2sets	2 sets	¥	900,000
4.12.	Paper filter(4 sets as spare)	5 sets	¥	1,000,000

Sub Total:	[4]	₩	52,900,000
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5 Supervisor fee:

- 5.1. M/C and gantry loader install
- 5.2. Process inspection
-7kinds X 7days
- 5.3. M/C training
- Operator : 5days
- Maintenance(mechanic+electric) team : 3days
- Translation(Marathi language) : 8m/d
- 5.4. Mass product support(included business trip 2days)
- 5.5. Hotel & meal with Domestic traffic fee & incidental fee
- 5.6. Air ticket
(T/O+training : 2man, Production support : 1man)

* Supervisor is dispatched by TPA ENGINEERING CORPORATION.

Sub Total:	[5]	₩	75,000,000
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6 Optional items

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| 6.1. | T/P shipping cost (INDIA<-> TPA) | 1 lot | ₩ | 5,000,000 |
| 6.2. | Measuring instrument and JIG shipping cost (INDIA<-> TPA) | 1 lot | ₩ | 1,500,000 |
| 6.3. | BIS certificate consultant fee(grinding machine + gantry loader) | 1 lot | ₩ | 60,000,000 |

Sub Total:	[6]	₩	66,500,000
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GRAND TOTAL: F.O.B. Busan port [A]	₩	660,950,000
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TPA ENGINEERING CORPORATION

[A] TOYODA CNC Angular-Wheelslide Cylindrical Grinder
model 60M GL5Ai-63 (wheel dia. of 610 mm, wide width max 125mm Specification)

* Specially designed and tooled up for
 finish-grinding of TRIPOD HOUSING

* Our specification No.

1	<u>Basic Machine</u>			
1.1	Basic machine with GC70S1 system model 60M GL5Ai-63	1 set	¥	174,120,000
2	<u>Extra Equipment:</u>			
2.1.	Live spindle workhead	1 set	¥	4,500,000
2.2.	Footstock, automatic(NC) reposition type, manual taper adjustment type - NC stroke 190mm	1 set	¥	6,500,000
2.3.	Rotary diamond roll type wheel dresser, mounted on the rear of workhead	1 set	¥	6,500,000
2.4.	Hydraulic oil pump unit, included low oil level warning switch with galvanizing(only internal of tank)	1 set	¥	5,600,000
2.5.	Lubricating oil pump unit, included low oil level warning switch with galvanizing(only internal of tank)	1 set	¥	6,600,000
2.6.	Paper filter type coolant supply equipment with magnetic separator, oil skimmer and ,bed cleaning pump, piping for cleaning bed, coolant cooler and sludge box. - Sludge box material : SUS - Apply galvanizing and anti-slip to top plate - Unable to galvanizing for the internal of tank	1 set	¥	40,000,000
2.7.	Standard service tool kit with torque wrench	1 set	¥	480,000
2.8.	Manual overhead open/shut type full cover and totally-closed cover	1 set	¥	8,000,000
2.9.	Air blowing unit	1 set	¥	3,500,000
2.10.	Piping for lubricating to center hole, R,L-side	1 set	¥	700,000
2.11.	Addition of 9kinds of workpiece design fee with programing fee(total 10 kinds)	9 kinds	¥	27,000,000
2.12.	2 gripper type gantry loader with Bypass function - Fanuc 35i series(with 2years maintenance contract) - Gripper unit - Drive gear / Helical Gear type rack & pinion drive - Gantry oil pan with galvanizing - Excluded air blow	1 set	¥	120,000,000
		Sub Total:	[1]+[2]	¥ 403,500,000
3	<u>OPTIONAL ACCESSORIES</u>			
3.1.	Automatic electric in-process gauging system, with 1 pce. of "MARPOSS "-make table-mounted type hydraulically-operated gauging head and control unit , pipings, wirings and fittings (UNIMAR S26, AMP P7), including automatic lateral locator, and auto wheel balacer	1 set	¥	52,000,000

3.2.	Maser bar for M/C alignment	1 set	₩	850,000
3.3.	Master work(using by WIA work)	0 pcs.	Deleted	
3.4.	3-steps signal tower with buzzer	1 set	₩	300,000
3.5.	Cooler for electric control box	1 set	₩	3,000,000
3.6.	Local lighting unit	1 set	₩	250,000
3.7.	Lighting units inside control box	1 set	₩	250,000
3.8.	Jib crane with Elec. type hoist	1 set	₩	2,500,000
3.9.	Air gun & coolant gun	1 pce.	₩	200,000
3.10.	Mist collector attaching arrangement (excluding mist collector)	1 set	₩	1,000,000
3.11.	Terminal box between loader and machine	1 set	₩	700,000
3.12.	Wheel guard auto offset device	1 set	₩	3,000,000
3.13.	Master bar for M/C accuracy check			DELETED
3.14.	Extra application to machine and equipment for KCs standard - noise filter, high efficiency motor, etc.			DELETED
3.15.	X-axis retract function for power supply off	1 set	₩	3,000,000
3.16.	Wheel trolley--two wheel mount type			DELETED
3.17.	Wheel balance stand & arbor	1 set	₩	5,000,000
3.18.	UPS system for power failure preparation			DELETED
Sub Total: [3]			₩	72,050,000
4	<u>Durable tool:</u>			
4.1	Grinding wheel flange(no spare)	7 sets	₩	40,600,000
4.2	Measuring head for auto locator(as spare)			DELETED
4.3	Measuring head for auto sizer(as spare)			DELETED
4.4	Feeler & contact for auto-locator(as spare)	1 sets	₩	1,500,000
4.5	Feeler & contact for auto-sizer(as spare)	1 sets	₩	3,000,000
4.6	Special type work driving center including driving plate (L-side only) (2 pcs. as spare)	3 pcs.	₩	6,000,000
4.7	Carbide-tipped tail stock center(R-side only)(2 pcs. as spare) - dead type center	3 pcs.	₩	1,500,000
4.8	Driving pin(for attaching drive center) - 5kinds X 2sets	10 sets	₩	2,500,000
4.9	Grinding wheel - 7kinds X 2sets	14 pcs.	₩	18,200,000
4.10.	Diamond roll(1 pce. as spare)	2 pcs.	₩	4,400,000
4.11.	Loader Jaw - 2kinds X 2sets	4 sets	₩	1,800,000
4.12.	Paper filter(4 sets as spare)	5 sets	₩	1,000,000

Sub Total:	[4]	₩	80,500,000
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5 Supervisor fee:

5.1.	M/C and gantry loader install	8	m/d	₩	8,000,000
5.2.	Process inspection -10kinds X 7days -2kinds X 1days	72	m/d	₩	72,000,000
5.3.	M/C training - Operator : 5days - Maintenance(mechanic+electric) team : 3days - Translation(Marathi language) : 8m/d	Included in outer race quotation			
5.4.	Mass product support(included business trip 2days)	Included in outer race quotation			
5.5.	Hotel & meal with Domestic traffic fee & incidental fee	80	lot	₩	20,000,000
5.6.	Air ticket (T/O+training : 2man, Production support : 1man)	Included in outer race quotation			

* Supervisor is dispatched by TPA ENGINEERING CORPORATION.

Sub Total:	[5]	₩	73,000,000
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GRAND TOTAL: F.O.B. Busan port [A]	₩	629,050,000
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